

PARTS LIST

Semiconductors:

- ESP32 module (SparkFun or Wroom)
- 8-pin, 3.65cm (1.44-inch) 128x128 coloured TFT display
- 6-pin SD card shield
- 8GB SD card

Miscellaneous:

- CON1 - 9-pin connector for ESP32 module
- CON2 - 8-pin connector for ILI9163 TFT display
- CON3 - 6-pin connector for SD card shield
- USB cable
 - 5V DC power supply
 - Jumper wires

and SD. Some library files are already inbuilt in Arduino IDE.

The sketch enables ESP32 to communicate with the TFT display and SD card. Then, it loads image files from the SD card to the loop. bmpDraw function in the sketch sizes the images and places these on the display, and the loop continues.

bmpDraw() function is available online from Adafruit and other TFT package suppliers for loading picture files (.bmp) to the SD card.

The header file (TFT_ILI9163C.h) for ESP32 was modified because images were not appearing on full screen.

ILI9163 TFT may have problems with full 128x128 resolution display. However, by tweaking TFT_ILI9163C.h file for the offsets variable, the problem can be resolved. To resize the images, use resize command on Ubuntu. The command is: convert -resize 128x128 source.xyz destination.jpg

Image files can be in any format, including .bmp and .jpg. Dump these in the SD card root directory as 1.bmp, 2.bmp, 3.bmp and so on. To convert picture files from .jpg or other formats to .bmp, use utility software like GIMP, PicsArt and MSPaint.

When you first switch on the power supply, the SD card may fail

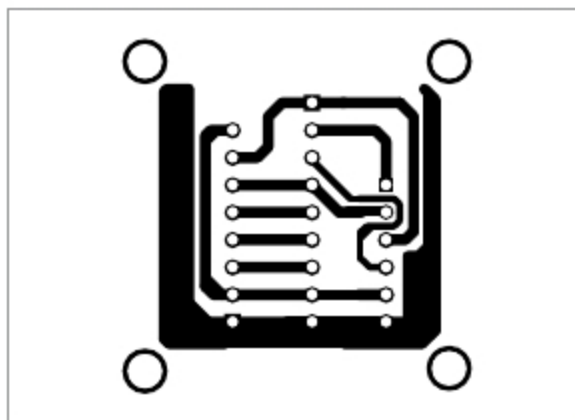


Fig. 4: Actual-size PCB layout of ESP32 picture slideshow

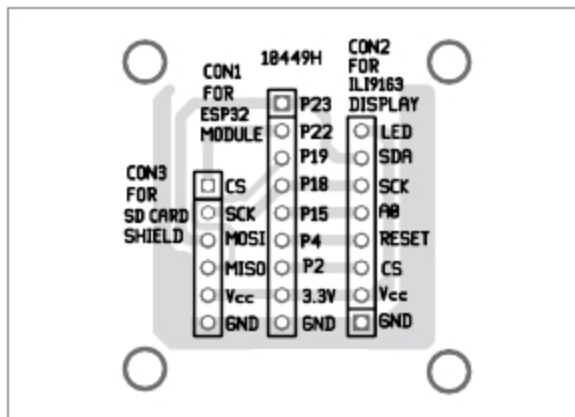


Fig. 5: Components layout for the PCB

to load. Restart ESP32 to solve the problem.

This project shows that display-related projects can be made easily on ESP32 rather than Raspberry Pi.

Construction and testing

An actual-size PCB layout for the picture slideshow is shown in Fig. 4 and its components layout in Fig. 5. Assemble the components on the PCB, as per the circuit diagram.

The connections are easy. Use appropriate ESP32 pins. In this project, all data pin connections of ESP32 board are located on one side.

In the prototype, a double-sided, general-purpose PCB was used. ESP32 was soldered on one side, while SD card shield and TFT display on the other.

SD card should contain 128x128 pixel .bmp images, so that these can appear perfectly on the TFT display. Number of images to be displayed depends on the memory capacity of the SD card. **EFY**



Somnath Bera is an avid user of open source software. Professionally, he is a thermal power expert and works as additional general manager at NTPC Ltd

EFY Note

The source code of this project is available for free download at source.efymag.com



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-Morris Hite

